MIL-G-52321A(ME)
28 May 1971
SUPERSEDING
MIL-T-52321(MO)
25 May 1964

MILITARY SPECIFICATION

GIN-POLE ASSEMBLY: CABLEWAY, MEDIUM

1. SCOPE

1.1 This specification covers a triangular-shaped, steel, gin-pole assembly used in the medium cableway.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-781	- Strapping, Steel, Flat and Seals.
PPP-C-650	- Crates, Wood, Open and Covered.

Military

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

FSC 5420

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DRAWINGS

ME

DL13208E5014 - Gin-Pole Assembly: Cableway, Medium.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY

A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.18 - Mild Steel Electrodes for Gas Metal-Arc Welding.
B3.0 - Standard Qualification Procedure.
D2.0 - Welded Highway and Railway Bridges.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, D. C. 20036.)

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3. REQUIREMENTS

3.1 Description. The gin-pole assembly or gin-pole sections, as specified, shall be in accordance with Data List DL13208E5014 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish a gin-pole assembly or gin-pole section, as specified, for examination and testing within the time frame specified to prove prior to starting production that his production methods will produce gin-pole assemblies or gin-pole sections as applicable, that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Welding electrodes. Welding electrodes for structural steel shall conform to AWS A5.1 or A5.18, E60XX series or E70XX series.

3.5 Identification marking. The sections of the gin-pole assembly shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.6 Cleaning, treatment, and painting. The sections of the gin-pole assembly shall be cleaned, treated, and painted as specified on the drawings except that the color number designation shall be 24087 or 34087 in accordance with FED. STD. No. 595.

3.7 Stenciling. The nomenclature, Federal stock number, weight, and cube shall be stenciled on each gin-pole section. The top section shall be stenciled on the top plate, the bottom section shall be stenciled on the base plate, and the center sections shall be stenciled on one of the

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vertical legs. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 3/4-inch high for top and bottom sections and 3/8-inch high for center sections. Stenciling shall conform to MIL-T-704.

3.8 Workmanship.

3.8.1 Steel fabrication. Steel used in the fabrication of the gin-pole sections shall be free from laminations, kinks, and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting with a tip suitable for the thickness of metal may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. All bends shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. All burrs, rough edges, and edges of holes shall be ground smooth.

3.8.2 Bolted connections. Boltholes shall be drilled undersize prior to assembly and reamed to full size after all welding and machining has been completed or shall be drilled to size after all welding and machining has been completed. All holes shall be drilled or reamed through steel templates. Boltholes shall be drilled or reamed straight through from one side of the component and shall be at right angles to the drilled surface of the component.

3.8.3 Welders and welding.

3.8.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society (B3.0).
Welding Qualifications of the ASME Boiler and Pressure Vessel Code (Section IX).

3.8.3.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall

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conform to the provisions of the American Welding Society publication "D2.0 - Welded Highway and Railway Bridges". All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

3.8.3.3 Jigs and fixtures. Components, except for minor parts, shall be assembled in steel jigs or frames and welded while securely held in position. Minor parts may be held securely and tightly in position by any means that will insure secure and correct positioning. The jigs or frames shall be designed so that distortion of the parts will be held to a minimum.

3.8.4 Interchangeability. Components of like description shall be fabricated to be duplicates within the tolerances specified and shall be interchangeable.

3.8.5 Assembly. The sections of the gin-pole assembly shall be capable of being assembled into a gin-pole assembly and erected, and there shall be no misfits or interference of mating parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance or rejection (see 6.4 and 6.5).

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4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction gin-pole assembly or gin-pole sections shall be examined in accordance with 4.4.2.1 and for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The gin-pole assembly shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection. When individual gin-pole sections are procured separately, the test shall not apply.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Individual. Each gin-pole assembly or gin-pole section, as applicable, shall be inspected for conformance to the dimensions covered by the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings under "Major - 100 percent inspection". Nonconformance to any of these dimensions shall constitute failure of this examination and shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The gin-pole assembly or gin-pole section shall be examined for the following defects:

Major

101. All dimensions designated major on the SQAP other than those under 100 percent inspection not as specified.
102. Material not as specified.

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103. Components missing or not as specified.
104. Workmanship not as specified.
105. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.

Minor

201. All dimensions not designated major on the SQAP not as specified.
202. Identification marking incorrect, missing, or illegible.
203. Treatment and painting not as specified.
204. Stenciling not as specified.

4.5.2 Assembly test. The gin-pole sections shall be assembled into a gin-pole assembly. The assembled gin-poles shall be erected and checked for assembly alinement and fit. Interference of mating members or incorrect fit at joints shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106. Disassembly not as specified.
107. Bolts and nuts not coated with preservative as specified.
108. Materials not as specified.
109. Crate not as specified.
110. Sections not banded together as specified.
111. Marking missing, illegible, incorrect or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation (see 6.6).

5.1.1 Disassembly. The gin-pole assembly shall be disassembled in three sections. Bolts and nuts involved shall be replaced and secured to prevent their loss.

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5.1.1.1 Bolts and nuts. Bolts and nuts shall be coated with type P-1 preservative conforming to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2 Packing. Packing shall be level A or C as specified.

5.2.1 Level A. Each complete gin-pole assembly, or gin-pole sections of like description when procured separately, shall be packed in a close-fitting crate conforming to PPP-C-650, style A, type II. Gin-pole sections procured separately shall be packed together in quantities not to exceed three each. Before packing in the crate, the sections shall be strapped together to form a compact bundle with three equally spaced straps conforming to QQ-S-781, type I or II, light duty, class B, grade 2, width 1/2-inch by not less than 0.015 inch thick. The bundled sections shall be secured within the crate in a manner to prevent free movement.

5.2.2 Level C. Complete gin-pole assemblies, or gin-pole sections when procured separately, shall be packed to assure carrier acceptance and safe delivery to destination at lowest rating in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.(see 6.6). Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The gin-pole assembly is intended to be used in erecting the towers of the medium cableway.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether complete gin-pole assemblies or top, middle, or bottom sections are required (see 3.1).
(c) Time frame required for submission of preproduction model (see 3.2).
(d) Level of packing required (see 5.2).

6.3 Preproduction model. Any changes or deviations of production gin-pole assemblies or gin-pole sections, from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish gin-pole assemblies or gin-pole sections conforming to this specification.

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6.4 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections. A suggested paragraph is as follows:

> "The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Definitions.

6.5.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 Levels of preservation have not been included as the preservation specified is the minimum acceptable for any level of protection. For purposes of preservation/packing level marking the marking shall be A/A, A/B, or A/C.

Custodian:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-G-52321A(ME) Gin-Pole Assembly: Cableway, Medium

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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